Pliocene – Pleistocene genesis for the Murray Brook and Heath Steele Au – Ag gossan ore deposits, New Brunswick, from paleomagnetism

D.T.A. Symons, M.T. Lewchuk, and D.R. Boyle

Abstract: Several Au–Ag gossans occur over massive sulphide deposits in the Ordovician Tetagouche Group near Bathurst, New Brunswick. The Murray Brook and Heath Steele B zone goethite gossans were about 45 and 15 m thick, respectively, prior to mining. They contain no minerals suitable for radiometric age dating. Geologically they must be younger than the Devonian Acadian orogeny and older than the last glaciation. Paleomagnetic methods were used to analyse specimens from 29 sites, mostly from ore on the pit walls. Host rocks and sulphide mineralization retain characteristic remanent magnetization directions in magnetite and pyrrhotite with a variety of directions that include possible Devonian overprints. Fertile and barren gossan specimens at 21 sites retain antiparallel normal and reversed A characteristic remanence components in goethite and (or) hematite. The A direction is $D = 357.7°$, $I = 61.7°$ ($\alpha_{95} = 4.5°$, $k = 31$). Its pole of 134.2°E, 85.2°N (dp = 5.4°, dm = 7.0°) falls on the North American apparent polar wander path and circumscribes the Earth's present rotational axis, indicating that the gossans formed during the Pliocene–Pleistocene. Examination of the site locations in the pits, along with the remanence polarity of their goethite and hematite A components, suggests the gossans formed during Chrons 1 and 2 only, or in the past 2.3 ± 0.3 Ma.

Résumé : Plusieurs chapeaux ferrugineux contenant Au et Ag apparaissent au-dessus des dépôts de sulfures massifs, dans le Groupe de Tétagouche, d'âge Ordovicien, près de Bathurst, Nouveau-Brunswick. Les chapeaux ferrugineux à goethite de la mine Murray Brook et de la zone B de la mine Heath Steele avaient une épaisseur de 45 et de 15 m, respectivement, avant le début de leur exploitation. Ils ne contiennent pas de minéraux appropriés aux datations radiométriques. Du point de vue géologique, leur origine doit être postérieure à l'orogenèse de l'Acadien au Dévonien et antérieure à la dernière englaciation. Les méthodes paléomagnétiques ont été utilisées pour analyser les spécimens prélevés sur 29 sites, généralement, à même le minerai sur les parois des trous d'excavation minière. Les roches encaissantes et la minéralisation de sulfures retiennent les directions d'aimantation rémanente caractéristique dans la magnétite et la pyrrhotite, avec une diversité de directions pouvant inclure des surimpressions dévoniennes. Les spécimens de 21 sites des chapeaux ferrugineux minéralisés, et stériles, retiennent les composantes normale antiparallèle et rémanente inversée A caractéristique dans la goethite et (ou) l'hématite. La direction A est $D = 357{,}7°$, $I = 61{,}7°$ ($\alpha_{95} = 4{,}5°$, $k = 31$). Son pôle fournit 134,2°E, 85,2°N (dp = 5,4°, dm = 7,0°), et il tombe sur la courbe de dérive apparente des pôles de l'Amérique du Nord et circonscrit l'axe de rotation actuel de la Terre, ce qui implique que ces chapeaux ferrugineux furent formés durant le Pliocène–Pléistocène. L'étude des localisations des sites dans les trous d'excavation minière et de la polarité de rémanence des composantes A de leur goethite et hématite suggère que ces chapeaux ferrugineux furent formés durant les chrons 1 et 2 seulement, ou durant les derniers 2,3 ± 0,3 Ma.
[Traduit par la rédaction]

Introduction

Eight Au–Ag precious metal gossans occur over massive sulphide deposits in the Ordovician Tetagouche Group of the Bathurst Mining Camp, New Brunswick (Boyle 1994). The Murray Brook gossan contained 1.7 Mt of ore at 1.21 g/t Au and 41.4 g/t Ag and the Heath Steele B zone gossan contained 0.2 Mt at 5.03 g/t Au and 172 g/t Ag (McCutcheon 1992; Fig. 1). Since radiometric dating of the gossan minerals is not possible, these gossans theoretically could be any age between the Devonian Acadian orogeny and the last glaciation, which began 40 000 years ago. Boyle (1991) estimated that it took at least 1 Ma under a constant hydrological and humid temperate climatic regime to form the deposits based on geochemical mass-balance arguments and conservative erosion rates. This paleomagnetic study was undertaken to date and to further constrain the genesis of these deposits.

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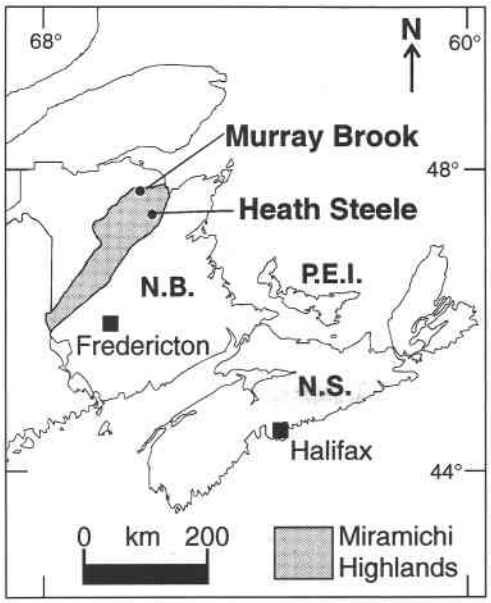
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Geology

The Tetagouche Group in the Miramichi Highlands of the Appalachians (Williams 1979; Fig. 1) is divided into six formations: the lowermost Vallee Lourdes limestones, Patrick Brook siltstones and quartzites, the Nepisiquit Falls tuffites, Flat Landing Brook silicic volcanics, Canoe Landing Lake basalts, and uppermost Boucher Brook shales and basalts (van Staal et al. 1992). These rocks yield Middle Ordovician

Fig. 1. Location map. Murray Brook is at 66.43°W, 47.53°N and Heath Steele B is at 66.04°W, 47.30°N.

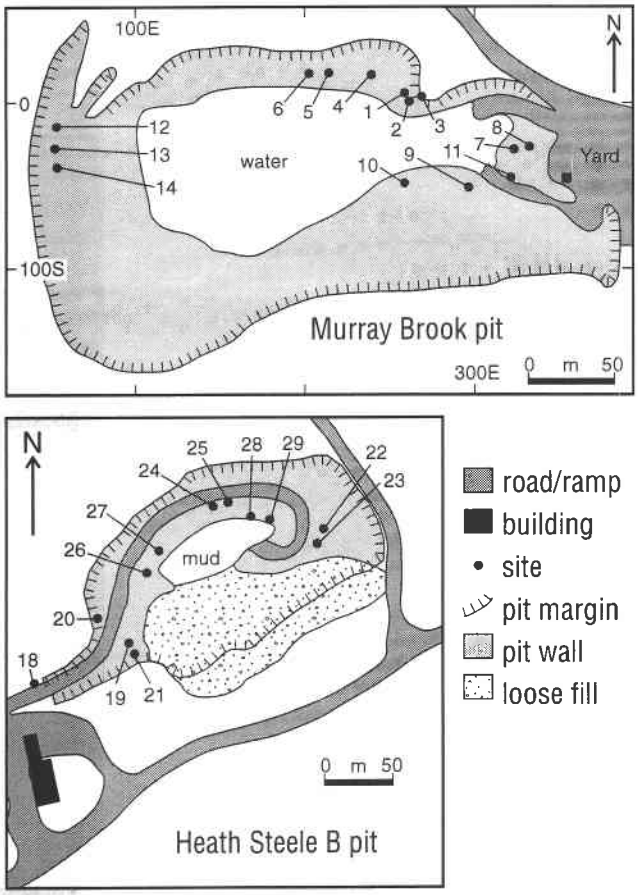


U–Pb zircon ages ranging from 471 ± 3 to 465 ± 2 Ma (Sullivan and van Staal 1990; van Staal et al. 1992). McCutcheon (1992) lists 37 massive sulphide Zn–Pb–Cu deposits with defined tonnages (including major mines), plus 58 other substantial occurrences in the Tetagouche Group of the Bathurst–Newcastle district. They comprise massive to disseminated pyrite, sphalerite, galena, chalcopyrite, magnetite, and, in places, pyrrhotite with trace to minor amounts of other sulphides. Sulphide reserves exceed 21.5 Mt at Murray Brook in the Patrick Brook Formation and 18.4 Mt in the Heath Steele B zones in the Nepisquit Falls Formation. The sulphides are thought to be primary (van Staal et al. 1992); however, conclusive evidence has been obliterated by later metamorphism (de Roo et al. 1991). These host rocks and ores have undergone four phases of later Paleozoic deformation to form isoclinal folds, and they have been regionally metamorphosed to greenschist facies (Helmstaedt 1971).

The gossan ores were developed over the steeply dipping sulphide deposits. At Murray Brook the gossan was up to 45 m thick before mining (Rennick and Burton 1992). Boyle and Burton (1990) mapped nine units, namely, about 3 m of overburden, hanging-wall Boucher Brook volcanics, barren gossan of no economic interest, fertile gossan with ore grades, gossanous breccia, quartz sand, pyrite sand, altered massive sulphides, and footwall Patrick Brook sediments. Fertile gossan contains goethite, euhedral quartz, secondary silica, plumbojarosite, jarosite, beudantite, cassiterite, and other minerals, including native gold and silver (Boyle 1994). Gossanous breccia defines preexisting fault zones. Quartz-sand units are thought to be the residuum of hanging-wall quartz–carbonate rocks. Pyrite sand marks the transition from gossan to massive sulphides. Altered massive sulphides show the conversion of chalcopyrite to chalcocite–covellite and the dissolution of carbonates and sphalerite.

At the Heath Steele B zone, the gossan ore was about 15 m thick. It comprised dark brown, massive goethite containing chalcopyrite boxworks (Dechow 1959). At the gossan's base was a 3 cm to 1 m thick layer of sandy dark material with acanthite or argentite, assaying up to 200 oz/ton (1 oz/ton = 34.3 g/t) in silver. Under this layer the oxidized ore contained friable, disaggregated pyrite; original interstitial pyrrhotite and chalcopyrite have been removed.

Fig. 2. Pit maps with site locations.



Experimental methods

Four or five oriented block samples were collected at each of 14 sites in the Murray Brook pit and 12 sites in the Heath Steele B pit, and five or six oriented cores were drilled at each of three sites near the Murray Brook pit (Fig. 2; Table 1). Of these sites, 19 are in gossan, five in altered wall rocks, two in unaltered massive sulphides, and three (the drilled sites) in unaltered host rocks. From 12 to 37 oriented specimens per site were prepared, for a total collection of 594 specimens.

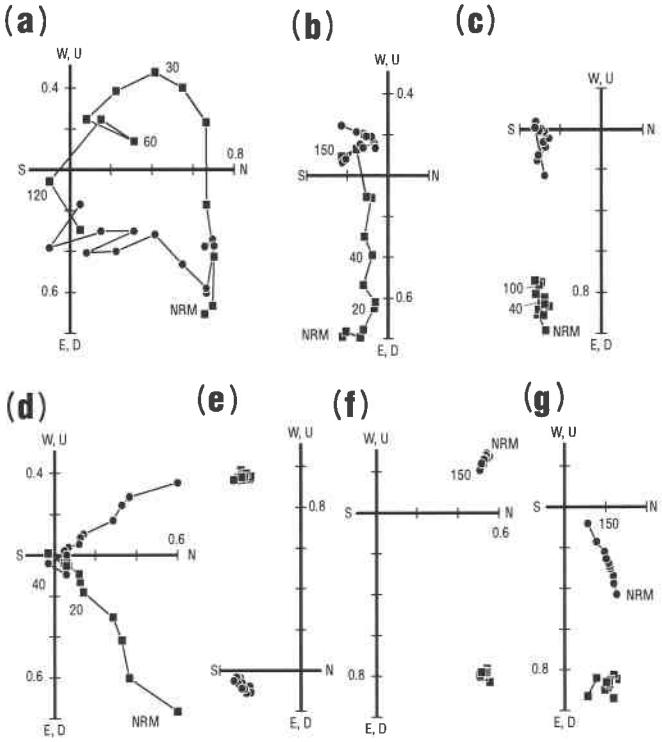
After 3 months' storage in a magnetically shielded room to allow viscous remanent magnetizations (VRM) to decay, the natural remanent magnetizations (NRM) of all specimens were measured on an automated CTF Systems Inc. DRM-420 cryogenic magnetometer. Twelve representative specimens were chosen from each site, usually three from each of the four blocks, except for site 11 with 15 specimens from five blocks. Two of these 12 specimens were thermally demagnetized at steps of 70, 85, 95, 105, 115, 125, 270, 295, 320, 500, 540, 570, 600, 560, and 680°C using a Magnetic Measurements TD1 unit to test for the diagnostic unblock-

Table 1. Remanence data for altered wall rock and gossan sites.

		Samples		Natural remanence							Goethite remanence							Hematite remanence						
Site	Lith	*B*	*S*	*n*	*r*	*d*	*D* (°)	*I* (°)	α_{95} (°)	*k*	*n*	*r*	*d*	*D* (°)	*I* (°)	α_{95} (°)	*k*	*n*	*r*	*d*	*D* (°)	*I* (°)	α_{95} (°)	*k*
1	fg	4	26	9	2	2	353	53	12	15	8	1	1	356	51	16	34	9	0	2	13	68	17	10
2	fg	4	28	12	0	0	0	63	7	40	12	0	0	359	62	9	23	9	0	0	10	70	30	4
3	afs	4	27	12	0	0	3	64	9	22	11	0	0	358	59	11	19	9	0	2	1	66	11	24
4	fg	4	20	9	3	0	173	58	23	5	0	4	7	173	−50	10	87	9	1	1	159	58	20	7
5	afs	4	19	3	7	1	161	−51	10	24	0	9	1	170	−44	12	20	5	4	1	331	54	13	16
6	fg	4	21	4	6	1	165	−54	7	48	3	6	1	166	−64	8	38	8	0	2	336	52	10	33
8	fg	4	17	12	0	0	41	49	27	4	8	1	2	39	62	21	7	10	0	2	17	75	15	12
9	fg	4	19	13	0	0	350	60	8	31	10	1	0	355	60	10	22	12	0	0	346	61	8	34
10	fg	4	16	12	0	0	10	54	7	36	11	0	0	5	51	7	38	9	0	2	29	64	8	46
11	bg	5	37	14	0	0	292	75	25	3	14	0	0	260	57	24	4	13	0	0	295	73	34	2
12	bg	4	16	9	2	1	48	66	23	5	9	2	0	60	70	23	5	4	0	7	37	74	6	234
13	bg	4	25	12	0	0	343	67	5	79	10	1	0	14	68	10	22	11	0	2	335	71	11	22
14	bg	4	22	9	0	1	328	65	6	80	7	0	2	335	65	7	82	3	0	6	333	69	14	57
18	fg	4	24	0	12	0	149	−65	14	13	3	6	2	143	−56	8	38	9	0	2	350	56	10	27
19	awr	4	19	3	9	0	190	−55	7	42	0	11	0	206	−53	23	5	2	6	3	180	−54	9	41
20	fg	4	7	3	7	0	185	−48	16	10	1	6	1	197	−45	24	8	1	6	3	179	−64	20	10
22	fg	4	19	9	3	0	357	55	9	22	11	0	0	350	60	7	46	3	4	4	172	−61	12	28
23	bg	4	26	11	1	0	9	51	13	12	7	0	4	16	38	10	37	9	0	2	51	41	12	18
25	bg	4	18	12	0	0	356	74	4	118	11	0	0	353	74	5	87	11	0	0	54	74	10	20
26	fg	4	21	9	0	4	348	55	17	10	6	4	2	344	62	14	13	8	0	4	336	148	19	9
27	fg	4	16	6	4	1	333	73	15	11	6	4	1	336	71	16	11	10	0	1	33	48	17	9
28	fg	4	13	10	1	0	150	63	21	6	10	0	0	144	64	25	5	10	0	0	156	64	18	8

Notes: *B*, blocks or cores collected; *S*, specimens with NRM directions; *n*, normal end points; *r*, reversed end points; *d*, deleted as aberrant; *D*, declination; *I*, inclination; α_{95}, radius of 95% confidence; *k*, precision parameter; Lith, lithology; afs, altered footwall sediments; awr, altered wall rocks; bg, barren gossan; fg, fertile gossan; fhs, "fresh" hanging-wall schist; fv, felsic volcanics; mv, mafic volcanics; so, sulphide ore.

Fig. 3. Normalized orthogonal plots of example specimens on AF step demagnetization. Projections on the horizontal (vertical) plane are shown by circles (triangles) with north–east–south–west (N, E, S, W) axes (up–down (U–D) axes). The steps are denoted in millitesla (mT). Axial intensities are expressed as a ratio of the NRM intensity. The rock types are (*a*) mafic volcanic (site 16); (*b* and *c*) "fresh" hanging-wall schist (sites 21 and 7, respectively); (*d*) sulphide ore (site 29); (*e* and *f*) gossan (sites 6 and 22, respectively); and (*g*) altered footwall sediments (site 3).



ing temperature ranges of goethite (75–125°C), pyrrhotite (270–320°C), magnetite (500–585°C), and hematite (600–675°C). One specimen per site at least was alternating field (AF) demagnetized at 14 steps to 150 mT using a Sapphire Instruments SI-4 unit to check for residual pyrrhotite or magnetite, which have coercivity spectra that preferentially demagnetize below 150 mT. Based on these tests, the remaining specimens for each site were demagnetized using five to seven thermal, AF, or both thermal and AF steps. Characteristic remanent magnetization (ChRM) directions were accepted if their maximum angular deviation (MAD) angle was $<10°$, with most $<3°$ (Kirschvink 1980). Multiple ChRM directions were determined for specimens in which multiple magnetic minerals were unblocked. Site and unit mean directions were calculated using Fisher (1953) statistics. Saturation isothermal remanent magnetization (SIRM) tests were run on 15 representative specimens using a Sapphire Instruments SI-6 pulse magnetizer by magnetizing them in 11 steps up to 900 mT, and then AF demagnetizing in 7 steps to 60 mT.

Results

Natural remanent magnetization

The NRM properties vary according to rock type. Massive sulphides have a median NRM intensity of 6.4×10^{-2} A/m, heavily altered wall rocks 9.9×10^{-3} A/m, gossan 7.9×10^{-3} A/m, and unaltered or slightly altered "fresh" host rocks 2.4×10^{-3} A/m. For each rock type, the intensities range about one order of magnitude on either side of the median. Their NRM directions are mostly either steeply down to the north or up to the south (Table 1).

Host rocks

Unaltered host rocks are effectively treated by both AF and thermal demagnetization. Most show the initial removal of steeply downward VRM near the present Earth's magnetic field (PEMF) direction, followed by a great-circle trajectory towards an underlying ChRM direction (Figs. 3*a*, 3*b*, 4*a*, 4*b*). Their intensities decrease smoothly up to just under 600°C, which is typical of titanomagnetite. The SIRM curves support this interpretation, showing rapid acquisition to near saturation by about 300 mT and rapid decay on AF demagnetization of the SIRM that are indicative of multidomain titanomagnetite (Fig. 5*a*, site 15).

Slight alteration of the host rocks could coincide with sulphide emplacement, and (or) weathering. AF demagnetization yields both minor and substantial intensity decay, but gives a ChRM without evident VRM (Fig. 3*c*). Thermal demagnetization provides evidence of mixed magnetic mineralogy. Some specimens show an initial ChRM in goethite, followed by a steady decrease in intensity that is typical of titanomagnetite with a different ChRM direction (Fig. 4*c*). SIRM analyses may also indicate two minerals, such as Fig. 5*a* (site 7), showing the rapid initial acquisition to 300 mT, which is typical of titanomagnetite, and then slow acquisition to 900 mT, which is indicative of goethite or hematite.

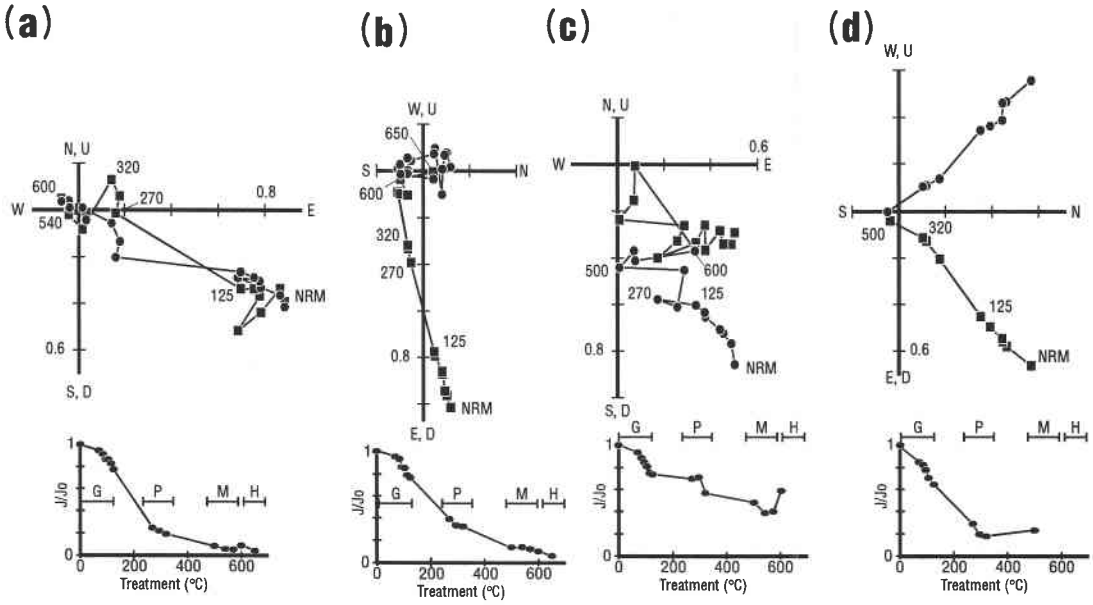
Massive sulphides

The massive sulphide specimens on AF demagnetization show the initial removal of a substantial steeply downward VRM (Fig. 3*d*) that is usually close to the PEMF direction. Over about 30 mT in the last few percent of intensity, some specimens begin a great-circle trajectory towards a shallower paleoinclination, suggesting that they may retain an underlying ChRM direction. The sulphide specimens show a consistent direction up to ~400°C that is preferentially unblocked in the pyrrhotite range (Fig. 4*d*). Above ~400°C, the sulphides oxidize to form new magnetic minerals, mostly magnetite, with increased intensity and aberrant directions. The SIRM acquisition curve shows a very rapid increase to saturation by about 200 mT and a decay curve that is indicative of multidomain pyrrhotite (Fig. 5*a*, site 29).

Gossan and heavily altered host rocks

The gossan specimens show less than a 10% intensity decay by 150 mT and virtually unchanged direction (Fig. 3*e*). Heavily altered host rocks similarly show negligible decay in most cases, although a few show evidence of residual unaltered primary magnetic minerals on AF demagnetization (Figs. 3*f*, 3*g*). Most gossan and heavily altered host rock specimens show a marked drop of remanence intensity below 120°C from goethite and a second marked drop to zero between 600 and 675°C from hematite. The ChRM in both minerals is usually directed steeply downwards to the north, the normal A component (A_N) (Figs. 6*a*–6*c*, 6*i*). The reversed A component (A_R) is found in both minerals also (Figs. 6*d*, 6*j*). A few specimens show the removal of A_R in

Fig. 4. Normalized orthogonal plot of example specimens on thermal step demagnetization with normalized intensity plotted beneath. Conventions as in Fig. 3 except steps are in degrees Celsius (°C). G, P, M, and H bars on the intensity plots define the diagnostic temperature range for demagnetization of goethite, pyrrhotite, magnetite, and hematite, respectively. The rock types are (*a*) mafic volcanic (site 15); (*b*) felsic volcanic (site 17); (*c*) "fresh" hanging-wall schist (site 7); and (*d*) sulphide ore (site 29).

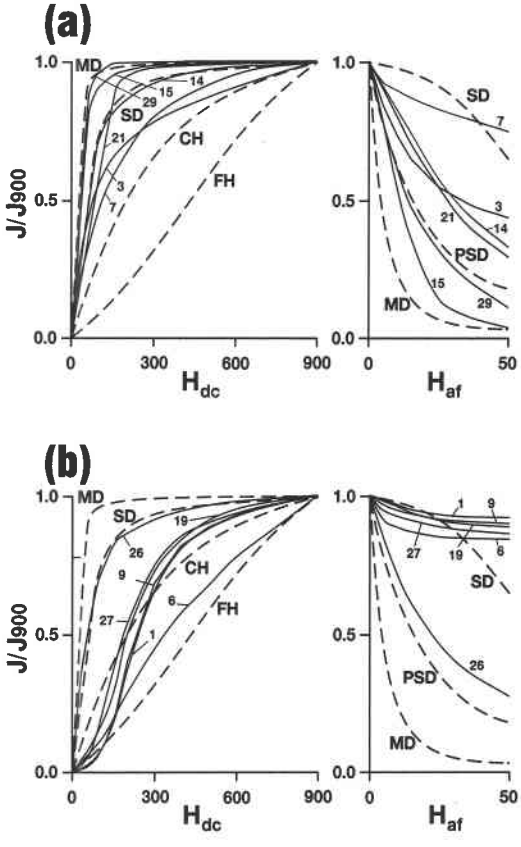


goethite, followed by A_N in hematite (Fig. 6*e*). In general, the remanence intensity of goethite per unit volume is several times weaker than that of hematite (Dekkers 1989), so goethite is obviously much more abundant than hematite. In some cases, only goethite is present (Figs. 6*f*, 6*k*); however, a few specimens have substantial hematite (Fig. 6*g*). Of particular interest are a few specimens that show the removal of A_R in goethite, followed by A_N in goethite (Fig. 6*h*).

Common in most of these specimens is a drop in remanence intensity between 270 and 320°C. Its explanation is uncertain. The 270, 295, and 320°C steps were put into the demagnetization sequence to check for residual pyrrhotite from unoxidized sulphides. However, Dekkers (1990) attributes similar drops for goethite between 260 and 360°C to its conversion to hematite. Why some specimens undergo very substantial drops whereas others show negligible drops is not readily explained by this mechanism, nor is the fact that some specimens show a final drop at ~300°C with no residual hematite. Another possibility is that trace amounts of greigite are present. It is an intermediate mineral in the transition from pyrite to goethite, has an intense magnetization comparable to magnetite, and is demagnetized between 260 and 340°C (Hoffman 1992; Reynolds et al. 1994).

SIRM analysis of gossan samples (Fig. 5*b*) evinces mixed mineralogy. Most acquisition curves lie between the single-domain magnetite and coarse-grained hematite type curves, probably reflecting goethite with a trace of magnetite, pyrrhotite, or greigite (Fig. 5*b*, sites 1, 9, 19, 27). Other curves are typical of hematite only (Fig. 5*b*, site 6) and of pyrrhotite with minor goethite or hematite (Fig. 5*b*, site 26). The heavily altered host rock SIRM curves are variable, showing initial rapid acquisition to 200 mT typical of pyrrhotite or magnetite and then a slower increase to 900 mT typical of goethite or hematite (Fig. 5*a*, sites 3, 14).

Fig. 5. Normalized isothermal remanent magnetization showing intensity acquisition (J/J_{900}) in a direct field in steps up to 900 mT, followed by AF step demagnetization of J_{900} up to 60 mT, for (*a*) non-ore specimens and (*b*) fertile gossan specimens. The curve numbers denote the specimen site.



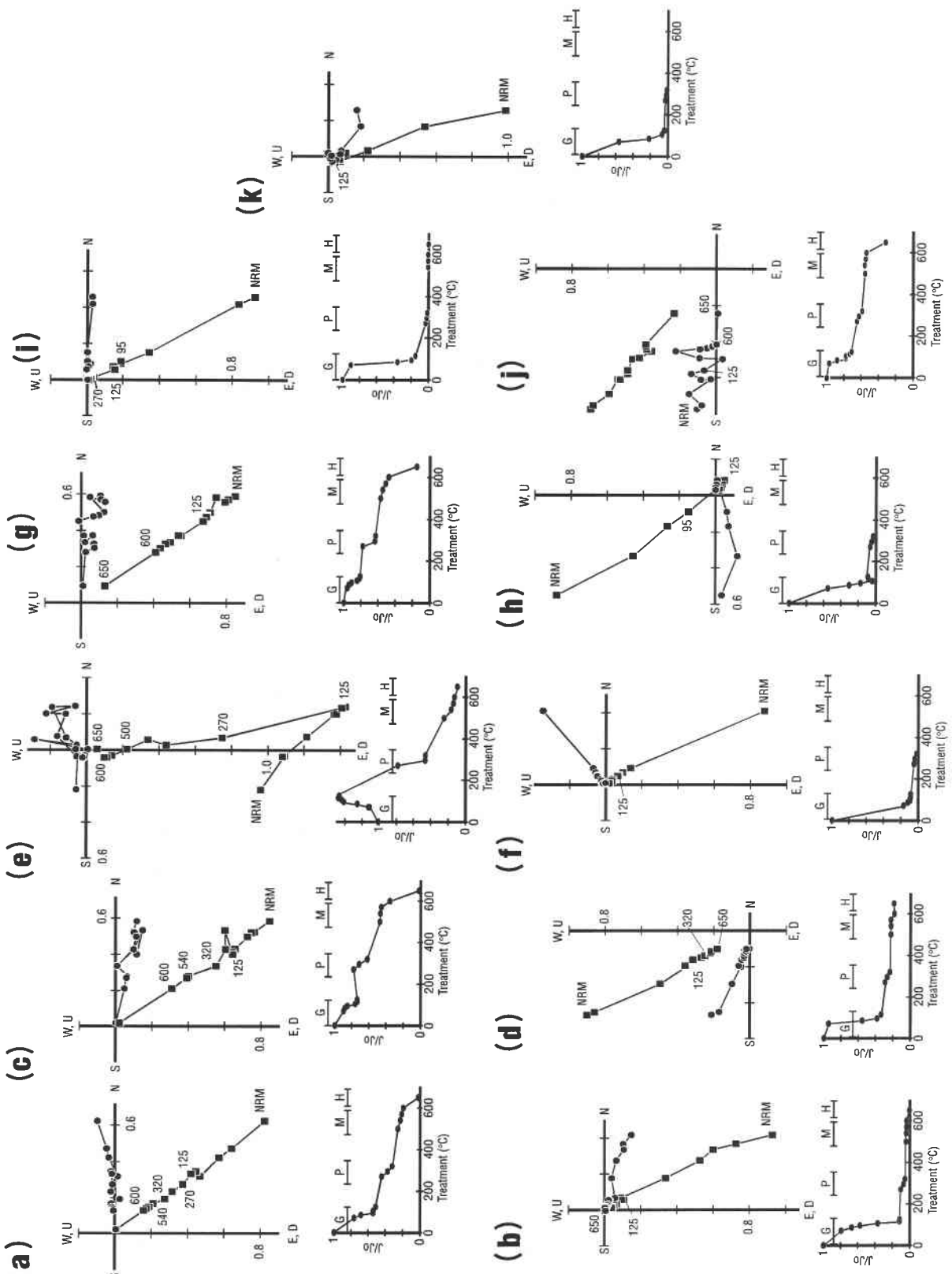


Fig. 6. Normalized orthogonal plots showing thermal step demagnetization of fertile gossan (*a*–*h*) and altered host rock (*i*–*k*) specimens from sites: (*a*) 1; (*b*) 9; (*c*) 11; (*d*) 20; (*e*) 4; (*f*) 14; (*g*) 8; (*h*) 26; (*i*) 22; (*j*) 19; and (*k*) 3.

Table 2. Remanence data for host rock and massive sulphide sites.

		Number					Demagnetizing		Direction			
Site	Lith	*B*	*S*	*n*	*r*	*d*	mT	°C	*D* (°)	*I* (°)	α_{95} (°)	*k*
7 a	fhs	4	20	6	0	0	0–120	70–570	178	52	22	10
b				6	0	0	25–80	70–540	88	18	19	10
15	mv	5	12	3	3	5	15–60	270–540	57	55	16	18
16	mv	6	17	0	13	0	20–90	270–320	53	−28	6	47
17	fv	6	12	12	0	0	15–90	270–540	197	73	10	21
21	fhs	4	21	7	0	4	20–100	125–370	193	61	19	11
24	so	4	22	12	0	0	20–60	85–320	302	78	11	16
29 l	so	4	24	6	0	6	15–90	85–320	304	49	11	33
h				6	1	5	60–90	500–540	284	−5	26	7

Notes: Abbreviations as in Table 1. Under "Site," a and b are two distinct components of comparable coercivity and blocking temperature, whereas l and h indicate lower and higher coercivity and blocking temperature components in the site.

Discussion

Host rock and massive sulphides

Both mafic volcanic sites give well-clustered but different ChRM directions (Fig. 7*a*; Table 2). The felsic volcanic site, in contrast, is relatively poorly clustered and gives yet another direction. These three sites are a few kilometres from Murray Brook pit. The two sites of "fresh" hanging-wall schist, one from each pit, yield poorly grouped ChRM directions, with site 7 yielding two populations of directions with comparable unblocking temperature and coercivity ranges. Both massive sulphide sites from the Heath Steele pit provide coherent clusters of ChRM directions to the northwest in the pyrrhotite unblocking temperature and coercivity ranges. At site 29 a second loosely clustered higher temperature and coercivity component is seen. Significantly, none of these sites defines an A_N or A_R direction. Thus the formation of the gossan with the A component postdates all folding, metamorphism, and sulphide mineralizing events, a conclusion readily apparent from the geology of the deposits. The modest cluster of site means formed by sites 7a, 17, and 21 gives a mean direction of $D = 187°$, $I = 62°$ ($A_{95} = 18°$, $k = 46$). Its pole position of 190°E, 1°S (dp = 22°, dm = 28°) overlaps the Devonian portion on the North American apparent polar wander (APW) path, thus providing suggestive evidence of an Acadian orogenic overprint or remagnetization of the host rocks.

Altered host rocks and gossans

For the altered host rocks and gossan sites, the mean NRM directions were first computed. Next, the vector-removed direction was determined by best-fitting the 20–125°C step directions to isolate the goethite ChRM component direction, the 270–320°C steps for the "pyrrhotite" component, and the ≥540°C steps for the hematite component. Nearly all specimens gave reliable directions for goethite and hematite, and their site means were computed (Table 1). All sites give consistent A_N or A_R mean directions for goethite and (or) hematite (Fig. 7*b*), except for site 28, which is always aberrant. Site 28 is located in what is now thought to be in a rotated slump block on the bottom of the Heath Steele pit.

The 21 site mean directions were averaged to get the unit

Fig. 7. Equal-area stereogram of site mean directions with down (up) vectors shown as filled (open) symbols for (*a*) mafic volcanic (triangle), felsic volcanic (inverted triangle), "fresh" hanging-wall schist (circle), and sulphide ore (square) sites as given in Table 2; and (*b*) goethite (triangle) and hematite (circle) ChRM components as given in Table 1.

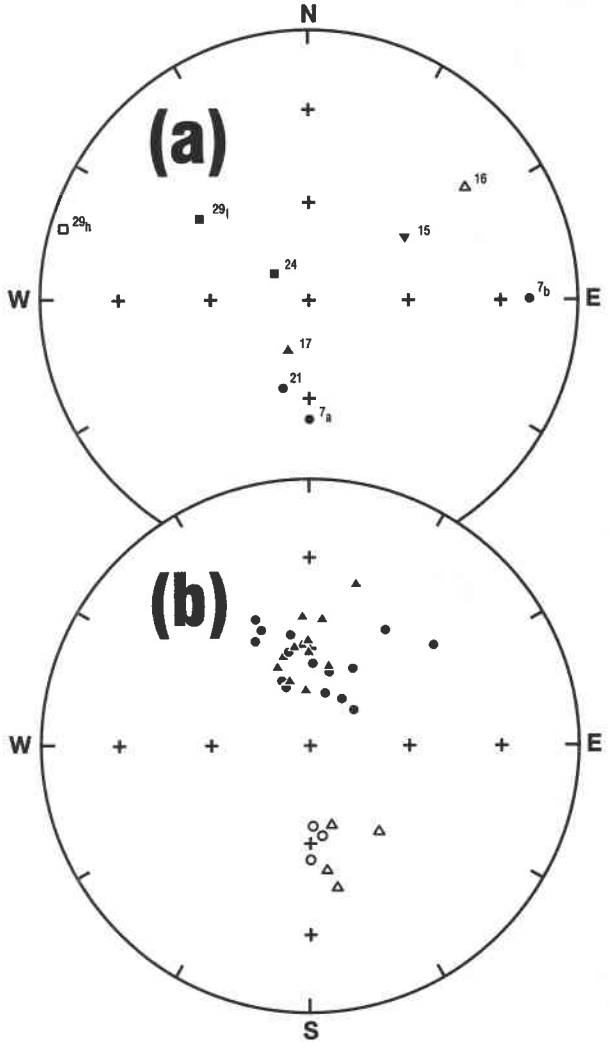
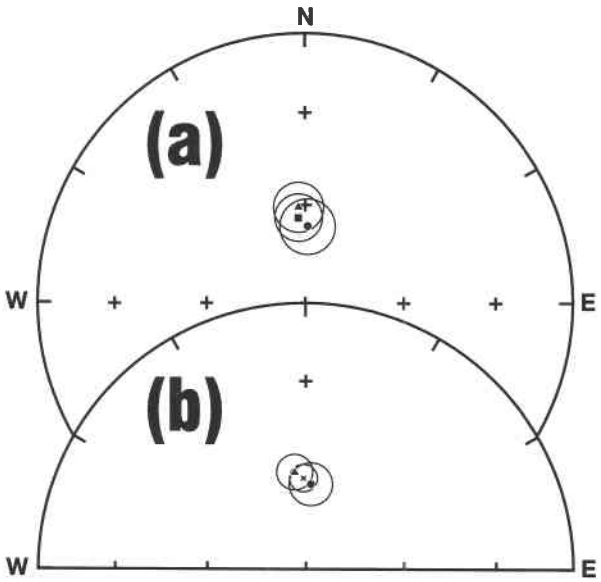


Table 3. Unit mean directions for altered wall rock and gossan samples.

Group	Excluded sites[a]	N	D (°)	I (°)	α_{95} (°)	k
NRM						
All sites		21[a]	357.1	64.2	7.5	19
Normal sites	4, 8, 11, 12	12	354.1	61.7	5.3	68
Reversed sites		5	171.4	−55.5	10.9	50
Combined	4, 8, 11, 12	17[a]	353.2	59.9	4.6	61
Goethite						
All sites		21[a]	357.4	61.3	6.9	22
Normal sites	8, 11, 12	12	356.9	60.7	6.3	48
Reversed sites	19, 20	4	163.8	−54.1	13.4	48
Combined	8, 11, 12, 19, 20(a)	16[a]	353.1	59.2	5.5	45
Hematite						
All sites		21[a]	2.3	67.4	8.0	17
Normal sites	2, 4, 11	15	3.7	64.6	8.5	21
Reversed sites		3	176.3	−59.7	8.5	210
Combined	2, 4, 11(a)	18[a]	2.4	63.8	7.1	25
Overall	Those in (a) above	34	357.7	61.7	4.5	31

Notes: N, number of sites, other abbreviations as in Table 1.
[a]Site 28 excluded also because of suspected slump-block rotation.

Fig. 8. Equal-area stereogram of the unit mean directions for altered host rocks and gossan sites showing the averages for (*a*) all 21 NRM (square), goethite (triangle), and hematite (circle) ChRM site means; and (*b*) the reliable ($k > 8$) goethite (triangle), hematite (circle), and goethite plus hematite (×) site means with their cones of 95% confidence (Fisher 1953).



mean directions (Table 3). The high stability of the remanence can be seen by comparing the NRM, goethite, and hematite means for all 21 sites that are the same at the 95% confidence level (McFadden and Lowes 1981; Fig. 8*a*). Using only well-defined site means with a precision parameter (k) (Fisher 1953) of >8, the means were obtained for the A_N, A_R, and combined-polarities A unit means (Table 3). Again the normal, reversed, and combined unit means are statistically the same at the 95% confidence level for the NRM, goethite, and hematite components. This result shows that these specimens do not retain a significant VRM component.

For both goethite and hematite, the A_R means have slightly shallower inclinations by ~7 and ~5° than the A_N means, constituting a class B and a class A paleomagnetic reversal test of McFadden and McElhinny (1990), respectively. This favours the preservation of a single ChRM component excluding polarity. The ~6° difference is not statistically significant, in part because there are relatively few reversed sites causing A_R to have larger radius for its cone of 95% confidence (α_{95}) (Fisher 1953). Nevertheless, the A_N and antiparallel A_R populations may be validly combined to get the A population average directions for goethite and hematite.

The goethite and hematite were formed in the same weathering event. Although their component directions come from the same set of samples, they represent radically different and nonoverlapping blocking temperature spectra. Thus, for calculating a pole position to date the event, it is reasonable to treat them as independant data sets. Since the combined goethite and combined hematite CRM directions are not different at the 95% confidence level (Fig. 8*b*, Table 3), the two sets may be drawn from one population with an overall average of $D = 357.7°$, $I = 61.7°$ ($N = 34$, $\alpha_{95} = 4.5°$, $k = 31$).

The dehydration component

The ChRM directions of 22 specimens could be defined in the 270−320°C interval. They give a mean direction of $D = 351.2°$, $I = 62.9°$ ($N = 22$, $\alpha_{95} = 15.6°$, $k = 5$) after switching polarity of the three reverse ChRM directions. Excluding eight ChRM directions that deviate by $>40°$, the central cluster of 14 vectors gives a direction of $D = 343.8°$, $I = 61.4°$ ($N = 14$, $\alpha_{95} = 9.4°$, $k = 19$). Both the overall and cluster of 14 directions are statistically similar to each other and to the goethite and hematite mean directions. This shows that this is not a primary pyrrhotite component. It suggests that the ChRM simply records the goethite−hematite

Fig. 9. Pole positions for the goethite (triangle), hematite (circle), and goethite plus hematite (■) gossan components with their ovals of 95% confidence in relation to the apparent polar wander path for the Mesozoic and Cenozoic (Van der Voo 1990, 1993), the far-sided biased pole for the past 5 Ma (×) (Merrill and McElhinny 1983), and the present Earth's magnetic field (PEMF).

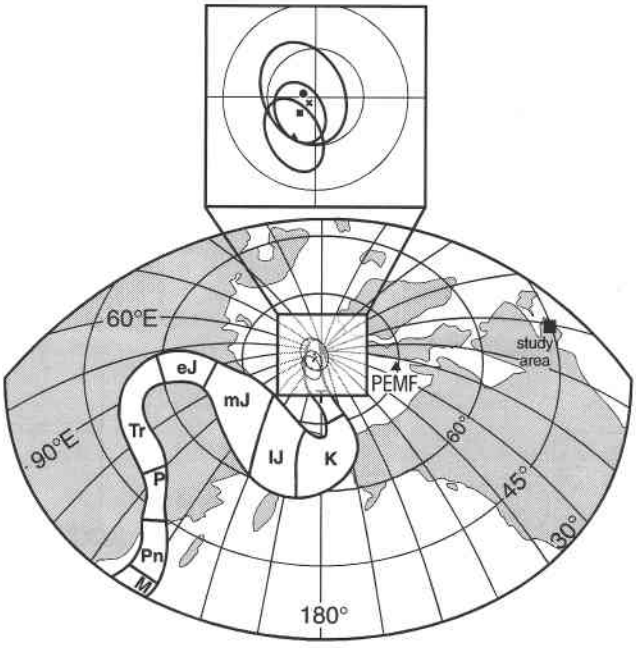
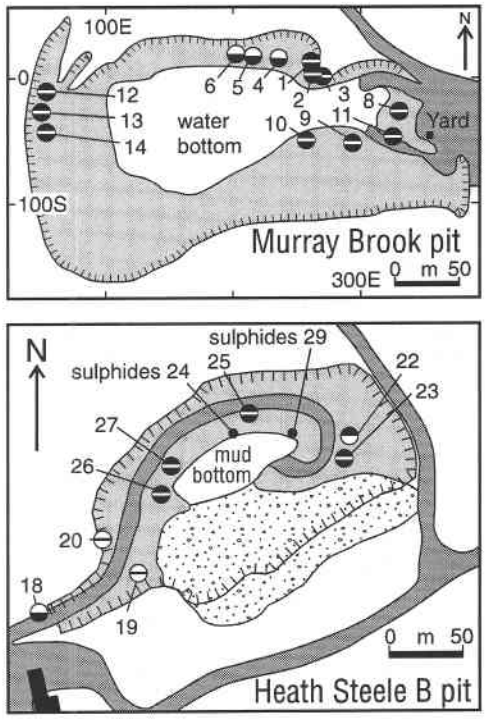


Fig. 10. Polarity of ChRM of altered host rock and gossan sites in the mine pits. The upper half of each circle gives the goethite polarity and the lower half the hematite polarity for the site. Filled and open semicircles indicate normal and reverse polarity, respectively.



transition upon dehydration, but does not rule out entirely that greigite is present as an intermediate product of pyrite oxidation.

Pole positions

The combined unit mean direction for goethite gives a pole position of 150.0°E, 81.9°N (dp = 6.2°, dm = 8.2°, N = 16), for hematite 72.9°E, 87.4°N (dp = 9.0°, dm = 11.3°), and for both together 134.2°E, 85.2°N (dp = 5.4°, dm = 7.0°) (Fig. 9). The overall pole falls directly on the Earth's present rotational pole or time-averaged magnetic field direction at the 95% confidence level.

Paleomagnetic data from several thousand sites worldwide for rock units of ≤5 Ma age give paleomagnetic poles for normal and reversed sites that are about 2 and 4° far-sided, respectively (Merrill and McElhinny 1983). Whether this bias is caused by a characteristic offset of the Earth's magnetic field, a geologic process such as sediment compaction, or faulty paleomagnetic procedures such as inadequate demagnetization is not clear. Noting that the overall gossan pole is about 5° far-sided, correcting for about 3° of far-sided bias brings it to within about 2° of the Earth's pole for the past 5 Ma, which is very much less than the α_{95} value of 4.5°. This comparison indicates that the ChRM of the gossan was acquired during the past 5 Ma.

Reversals

The ChRM polarity pattern found in the gossan sites further constrains the age of ore genesis. Two reasonable assumptions are made. First, following the reaction paths shown by Eh−pH diagrams for oxidation (Garrels and Christ 1965), it is assumed that the primary sulphides are progressively oxidized to goethite and they then dehydrate to hematite, and that any magnetite is oxidized to hematite coincidentally. The second assumption is that downward-circulating meteoric groundwater was the oxidizing agent, which matches the present hydrologic regime. Thus, remanence acquisition proceeds from surface downwards as the oxidizing front forms the gossan. In this model, the youngest remanence would be forming at present in goethite at the base of the gossan atop unaltered sulphides, as is seen in the pits, and the oldest remanence would be held by surficial goethite that has escaped conversion to hematite.

At Murray Brook, 10 of 13 sites retain a normal polarity in both goethite and hematite (Fig. 10). They include sites next to primary sulphides at the bottom (sites 1−3, 8−10) and end (sites 12−14) of the pit. The model interpretation is that these gossan sites acquired their remanence during the Bruhnes normal epoch or polarity Chron 1n, which extends back to 0.78 Ma (Cande and Kent 1992). The remaining three sites (sites 4−6) are farthest from unoxidized ore and retain a reversed remanence in goethite and a normal remanence in hematite. For these, it is likely that goethite was formed during the preceding reversed Chron 1r and that some of it was dehydrated and converted to hematite during the subsequent Chron 1n. This suggests that the gossan formed in about 1 Ma. Based on mass-balance calculations for gold, Boyle (1991) argued that a volume equal to the mined-out gossan was eroded before glaciation. This suggests that the entire gossan was formed over the last 2 Ma.

At Heath Steele the four lowest sites (sites 23, 25–27) at the bottom of the B pit next to primary sulphides retain a normal polarity in both goethite and hematite that was formed during the Chron 1n (Fig. 10). Halfway up the pit wall at sites 19 and 20, the gossan retains a reversed polarity in both goethite and hematite, implying that both minerals were formed during Chron 1r. Site 22, at about the same level in the pit wall, has a normal remanence in goethite but a reversed remanence in hematite. This implies that the goethite was formed during Chron 2n. The uppermost site, 18, has the opposite combination, with reverse polarity in goethite and normal polarity in hematite. Two options are tenable. Looking at adjacent sites 19 and 20, the probable explanation is that the goethite was formed during Chron 1r and some converted to hematite during Chron 1n. The alternative, suggested by site 18's elevation, is that the goethite was formed in Chron 2r and that some converted to goethite during either Chron 2n or 1n. Chron 2n extends back to 2.60 Ma, setting a maximum age for the gossan.

In summary, the polarity and mineralogy data for the gossans require a minimum time for the Earth's magnetic field to undergo a normal–reverse–normal sequence, ending with Chron 1n. An alternative explanation adds an initial reverse interval to the sequence. Thus the minimum age for the onset of gossan formation would be about 1.05 Ma, assuming Subchron 1r.1n was captured. The maximum age is more tenuous. If Subchron 1r.1n was not captured and the samples record the main polarity of each chron without omission, then the longer sequence of two reversed and two normal polarity chrons indicates the onset after 2.60 Ma. It is more likely, however, that a better estimate is 2.3 ± 0.3 Ma. Certainly the options of a rapidly formed postglacial gossan or, alternatively, an age in excess of 3 Ma are ruled out.

Conclusions

It is concluded that the Murray Brook and the Heath Steele auriferous gossan ores acquired their ChRM when primary Paleozoic sulphides were oxidized to goethite and minor hematite. Their overall dual-polarity ChRM direction is $D = 357.7°$, $I = 61.7°$ ($N = 34$, $\alpha_{95} = 4.5°$, $k = 31$). The mean of the ChRM directions defined in the gossan specimens for the 270–320°C blocking temperature spectra coincides with the overall direction. This records, most likely, the dehydration of a minor amount of goethite to hematite rather than residual primary pyrrhotite. Both host rocks and unoxidized primary sulphide ore give widely divergent ChRM directions, with some suggesting a late Paleozoic age. The gossans' overall pole position of 134.2°E, 85.2°N (dp = 5.4°, dm = 7.0°) is concordant with the pole for the North American craton during the last 5 Ma. Tracking the polarity pattern in goethite and hematite of the sites indicates that the gossans began to form at least 1.05 Ma ago and, more probably, at about 2.3 Ma. Thus these ore deposits were formed during the Pliocene and Pleistocene, beginning substantially before the last glaciation, and that they are not an exhumed paleogossan.

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